Bhavya Dosi

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PROFESSIONAL EXPERIENCE

VT Competitive Robotics Organization (CroLabs) **Sep 2025 - Present**
Chief Engineer *Blacksburg, Virginia*
HIVE - Heterogeneous Intelligent Vehicle Ensemble
- Led a team of 25+ engineers across software, electrical, mechanical, and drone operations to deliver HIVE, a platform that automatically recognized unfamiliar robots, inferred their capabilities, and coordinated them with a single plain-English command
- Built an onboarding pipeline that parses a robot's ROS 2 workspace, collects line-level interface evidence, and uses an LLM to generate a structured capability profile, cutting onboarding time from months to minutes
- Built the runtime those profiles drive: self-configuring navigation, flight, and manipulation skill libraries, a shared world model every robot reports into, and a planner that validates each step against it before dispatch

University Libraries at Virginia Tech **Oct 2025 - Present**
Technical Consultant *Blacksburg, Virginia*
- Enabled rapid prototyping by assisting students and researchers with 3D printers and laser cutters, using Fusion 360 to prepare printable models, which reduced prototype development time by up to two days
- Mentored and trained patrons on design-for-manufacturing principles and the operation of 3D printers, laser cutters, and Resin printing, leading to higher-quality prototypes and increased equipment usage
- Delivered technical consulting, troubleshooting complex CAD and hardware issues for patrons daily to accelerate project turnaround

LEADERSHIP EXPERIENCE

DigiKreate **Oct 2022 - May 2025**
Founder & CEO
- Founded a design startup leveraging AI tools to automate workflows, managing 20+ clients.
- Managed client projects and utilized generative AI to optimize the design process for efficiency and scale.

PROJECT EXPERIENCE

Jarvis AI- Agentic orchestrator
- Built an AI orchestrator that breaks a task into sub-tasks and routes each to the best-fit LLM (Claude, Gemini, GPT, local Ollama) by cost and complexity, with shared memory across model switches.
- Designed a plug-and-play skill system: drop-in Python tools are auto-discovered and exposed via MCP, with all agent-written code run in isolated E2B sandboxes.
- Built on modular microservices - LangGraph orchestration, LiteLLM / RouteLLM routing, Mem0 + ChromaDB memory, CrewAI developer/QA agents, and an Ollama offline fallback.

Multimodal Robot - Air & Land
- Co-developed an air/ground robot with PID-controlled actuators, custom PCBs, and laser-cut frames that transitions cleanly between all three environments
- Ran FEA/CFD analysis on how mechanical and hydrodynamic loads shift across air and ground to drive material and geometry decisions.

Natural-Language Robotic Arm
- Built an embodied AI agent that turns natural-language and voice commands into physical actions on a 7-DOF Franka Panda arm in MuJoCo simulation
- Engineered a perceive–reason–act loop - tool-calling language agent, NVIDIA NIM vision-language grounding, and differential IK with weld-based grasping.

EDUCATION

Virginia Tech **May 2029**
Mechanical Engineering
- **GPA:** 3.86
- **Achievements:** Dean's List w Distinction

SKILLS

Robotics & Autonomy: ROS 2, Gazebo, Nav2, MuJoCo, SLAM, Lidar Integration, Inverse Kinematics, PID Control, Sensor Fusion, Multi-Robot Coordination, Robotic Manipulation, Machine Learning, Computer Vision
Embodied & Agentic AI: AI Agents, Multi-Agent Orchestration, Model Context Protocol, Tool / Function Calling, LLM, VLM, VLA, World Models, Prompt Engineering, Loop Engineering
Programming & Software: Python, C++, Arduino, Docker, Git / GitHub, Linux / WSL2
Hardware: PCB Design, ESP32, micro-ROS, Microcontrollers, Sensors & Actuators, Raspberry Pi, IoT
Prototyping & CAD: CAD, Fusion 360, SolidWorks, URDF / Robot Modeling, 3D Printing, Laser Cutting, Rapid Prototyping